THE TRIPOD APPROACH
A SYSTEMATIC AND STRUCTURED FRAMEWORK TO DESIGNING APPLIED TEXTILES

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ABSTRACT

The term 'next to my skin' is frequently used as a way to describe the close bonds between human beings and textiles. It is also widely accepted within the broad field of textile professionals that textile attributes related to emotional value – for example hand, look and drape – can be a make-and-break decision for a company.

However textile design research has shown that experience of emotional value is closely connected to fabric perception, and much literature suggests that textiles are experienced by a combination of sensory evaluation and memory associations.

This paper introduces the Tripod Approach as a framework that can support designing the stimuli, which can lead to personal experiences of emotional value in relation to applied textiles. It can be used to access the relevant level of entry in the textile design process focusing on appropriate design parameters, and support designing the textile as part of an overall context.

Keywords: Textile design, emotional value, framework, systematic approach, empathy

EMOTIONAL VALUE OF APPLIED TEXTILES

The fields of philosophy, neuroscience and psychology offer many contradictory theories on emotion, and within design research there are several suggestions of how to address emotional aspects in design in general (for a brief overview see Bang, 2010).

Textile design research has shown that experience of emotional value is closely connected to fabric perception based on touch (Moody et al., 2001) and vision (Homlong, 2006). Gale & Kaur (2002) argue that lifestyle plays a significant role in our lives and therefore pleasure and well-being are central themes in modern culture and has to be considered by the textile industry. They also notice that textile attributes can be a make-and-break decision for a company. This is a viewpoint that is also shared within textile science (Hatch, 2006) and textbooks teaching aesthetics to design students (see for example Fiore & Kimle, 1997). A study of fabric perception shows that the characteristic 'feel' of a fabric depends on a combination of variables involving the surface texture and what is referred to as 'emotion/cognitive/mood associations' (Moody et al., 2001: 1). Gale and Kaur also suggest that textiles are experienced as a combination of sensory evaluation and memory associations.

Still the term 'emotional' is not a formalized professional term in textile design practice. This paper is concerned with the possibilities of accessing the 'emotional' design possibilities in industrial textile design. Therefore the next section will introduce textile design practice and the design process in an industrial context.

INDUSTRIAL TEXTILE DESIGN

Even though the present financial crisis has had a serious influence on the Danish textile industry, innovation is integrated in the strategies for how to handle the crisis (Bloch, 2008; Axholm, 2009). Uncertainty, crisis and chaos in the world paired with an increased industrial focus on competitiveness in a

globalized and knowledge-based economy motivates the designers to rethink and sharpen the design process.

On a practical level technical development and outsourcing means that all manufacturers in the textile industry in principle have equal access to developing and producing textiles with a high aesthetic, functional and technical quality. Although still of high importance technical specifications, such as color fastness to light and abrasion resistance, are no longer a unique sales parameter but a presumed precondition, since the customer expects high quality. For the manufacturer this means that the focus is further extended to include service, technological development, and function as well as the experiences, stories and emotions that are attached to the textiles.

THE TEXTILE DESIGN PRACTICE

Textbooks often refer to textile designers as designers within the textiles and clothing industry who work according to design briefs and intended end-use developing ideas and realizing concepts that can produce 'marketable fabrics' (Gale & Kaur, 2002: 37) and 'viable fabric designs' (Wilson, 2001: 10).

Textile designers working with industrial design can be in-house designers at manufacturers or they can work as independent freelance designers or through an agent (Wilson, 2001).

In-house designers can be involved in all stages of product development, process management, and customer and supplier contact on a regular basis, because they are familiar with the company. They can be involved in all processes from setting a design brief, through product development, and sale and marketing of the finished product (see for example Wilson, 2001).

Jacquie Wilson (2001) argues that traditionally the textile designers in the industry produce paper sketches and fabric samples developing designs for weave, print and knit to be mass-produced to a defined market. But as Colin Gale and Jasbin Kaur (2002) argue the textile design practice today is much broader:

"Most textile professionals acquire the complex range of knowledge and skills appropriate to their sector, piece by piece, building their expertise over years; few would claim to understand the full diversity of the textile world and its history" (Gale & Kaur, 2002: ix).

Textile designers are constantly experimenting with aesthetic, functional, and technical crossovers between techniques, finishing, surface treatments, and materials. In addition to weave, print and knit the techniques can for example be: crocheting, tufting, bonding, felting, embroidery, welding and laser cutting. Considerations about visual appearance, texture, structure, application, functionality, end-use, advanced technological materials, ethics and sustainability are also included in the design process (Gale & Kaur, 2002).

A COMPLEX AND NON-LINEAR DESIGN PROCESS

Josie Steed (2009) has conducted a survey of 12 Scottish textile enterprises involved in 'design and manufacture' partnerships, mapping the creative processes in order to get insight into design innovation. One of the conclusions of the study is that contemporary textile designers need to be 'hybrid' designers: "Designers need to be able to understand and work across disciplines" (ibid: 445). The British textile industry has changed and the designer needs to develop her skills accordingly in order to be able to experiment with new technology and market requirements.

This is similar to other design domains. Based on studies of how designers work and collaborate and through teaching design students from different design domains the British architect and psychologist Bryan Lawson (2006) argues that designers of today must be able to learn to appreciate and exploit new technology as it develops, and design students must realize that they need to learn more than a few traditional disciplines.

Donald Schön (1983) defines designing as 'having a conversation with the materials of a situation' and problem setting as a crucial activity in the design process. Reflection-in-action is a core concept in Schön's understanding of the design process and comparable to other researchers' definition of design

processes as highly complex processes where problem and solution emerge together (in general, see e.g. Lawson, 2006; in textile design, see e.g. Wilson, 2001).

Seen from the perspective of an in-house textile designer the organizational structure, the everyday practice and the textile design process are intertwined. The design process is just one of the processes in a development project. It may be influenced by and also influence processes in other internal business units dealing with issues such as quality, environment, sales, marketing and logistics. This means that several business units work parallel on various aspects of the project. The design process may also be influenced by and influence the customer's design and development processes. There are always several projects running simultaneously, and the design process is constantly interrupted or influenced by other projects, deadlines, other business processes, customers, suppliers, test runs etc. In everyday practice the industrial textile design process is a non-linear and complex design process embedded in an overall linear, structured, and formal development procedure.

Thus today's textile design is a broad profession, and the textile designer possesses and constantly adds to a diverse and complex range of skills dependent on her interests and job situation. What further characterizes the textile designer is also that she can go beyond the design of the product and get involved in designing the material to be used in the product instead of merely choosing it. In order to capture the multiple aspects of textile design practice as well as the complex design process I propose to adopt an analytical and systematic approach to designing applied textiles.

The tripod Approach is developed on the basis of an existing teaching tool (see Appendix A in Bang, 2010). The teaching tool will be introduced in the following section.

THE TEACHING TOOL

Previous experiences from teaching textile design students have shown that it can be relevant to structure a systematic and analytical approach to textiles according to three perspectives considering the distance to the textile as well as the character of the perception. The three perspectives can embrace the multiple levels from fiber to finished product in textile design: The first one, 'the structure – the surface' perspective, refers to surface appearance and structure, for instance the material properties, the technique, and the pattern. The second one, the 'as piece goods – as a whole' perspective, considers the piece of fabric as a whole or as part of an object, for instance drape and the fabric composition in relation to the object. The third perspective, the 'as interior – in use/applied' perspective, concerns the use and function of the product, for instance interactive and responsive attributes, genre, or style.

We saw that the systematic approach to textiles enabled the students to address and analyze the finished design or the design idea in a methodical fashion.

Each perspective has a narrow focus on the textile. The perspectives combined provide a holistic impression of the textile as part of an overall context.



Figure 1. It can be relevant to structure a systematic and analytical approach to textiles according to three perspectives. The three perspectives can embrace the multiple levels from fibre to finished product in textile design. The figure shows the same fabric perceived from three different textile perspectives. Left: 'fabric – textiles as material', a detailed view of the fabric from a 'short' distance. Middle: 'object – textiles as part of an object', the same piece of fabric on a lounge chair seen from a 'normal' distance'. Right: 'Textiles as part of an environment', the lounge chair in a use context from a 'longer' distance (based on Appendix A in Bang, 2010).

The figure above illustrates the three perspectives. Analyzing an existing textile product focusing on the fabric detail completely ignores the specific context: type of chair as well as the lounge setting. The same is happening when the chair is the focus: the environment as well as detailed information about the surface appearance and the fabric structure is left out of consideration. When the lounge environment is the focus of attention that perspective ignores the other views. Using the tool systematically moving from one

perspective to the next will add new layers of information to the analysis or the idea development and thus enhance the understanding of the actual or future piece of textile.

THE THEORETICAL FOUNDATION OF THE TRIPOD APPROACH

The three perspectives of textiles in the teaching tool are inspired by the theory of Alois Riegl and the development of the Tripod Approach is further supported by the theory of Juhani Pallasmaa. Both will be outlined in this section.

THE OBSERVANT PERSPECTIVE

The observant perspective of the teaching tool as well as the Tripod Approach are inspired by the Austrian art historian Alois Riegl's theory of perception as explained in a Danish translation of chapter one of Riegl's seminal work "Spätrömishe Kunstindustrie" from 1901 (Riegl, 1901). Riegl's concern was to understand antiquity architecture from the Egyptian pyramid to the Roman basilica on its own terms. He developed a theory of perception based on the observer's position in relation to the object and the character of the perception, and it is these aspects of his work that is applied to the Tripod Approach.

Riegl uses the German term 'Sicht' to describe the observer's position in relation to the object. 'Sicht' can be translated as 'view', referring to the distance to the object: near, normal and distant view. The character of the perception is defined as either haptic or optic. According to Riegl haptic as well as optic refer to visual perception. Therefore haptic perception is tied to tangibility and the sense of touch, experienced as the visual perception of, in this case, a building's two-dimensional surface. Optic perception refers to the sense of vision, and, in this case, the experience of the building as a part of an environment.

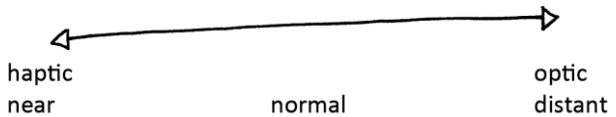


Figure 2. The figure shows Alois Riegl's three views – close, normal and distant – and the character of the perception, which can be haptic or optic.

The close view refers to a short distance to the object, which appears as a two-dimensional surface without form-giving shadows. The character of the perception at a short distance is what Riegl calls a haptic perception because it is possible to perceive surface details. The position of a normal view includes the surface as well as form-giving shadows. At a normal distance the perception can be both haptic and optic. Finally a distant view refers to a longer distance to the object where the shadows appear as dark areas without a form-giving effect. Seen from a longer distance the character of the perception is optic (Riegl, 1901).

THE PARTICIPANT PERSPECTIVE

It can be argued that taking an observer's standpoint, like Riegl suggests, may be too narrow and limiting in some respects. Therefore the relevance of incorporating a participant aspect in each of the three perspectives of textile design will be discussed in this section.

This section describe how the research of the Finnish architect and theorist Juhani Pallasmaa has inspired to further development of the systematic and analytical approach to textiles in this project (Pallasmaa, 2005). One of the cornerstones in Pallasmaa's work is that an architectural experience is much more than just a visual experience:

"An architectural work is not experienced as a series of isolated retinal pictures, but in its fully integrated material, embodied and spiritual essence. […] Unconscious peripheral perception transforms retinal gestalt into spatial and bodily experiences. Peripheral vision integrates us with space, while focused vision pushes us out of the space, making us mere spectators" (Pallasmaa, 2005: 12-13).

Pallasmaa acknowledges the importance of vision and argues that a problem occurs when vision is isolated from or suppresses other senses. He therefore proposes a multisensory and participatory approach to the experience of architecture. He emphasizes the importance of knowing the difference between a focused vision and conscious peripheral perception, between being a mere observer of the architecture and being a participant experiencing the architecture

with all senses. He warns that a one-sided focus on architecture as an isolated series of images may bias the architectural experience. The same approach seems to be appropriate when dealing with textiles and the experience of applied textiles. The figure below indicates the difference between observing an office chair and participating with all senses in the experience following Pallasmaa's idea of focused vision and peripheral perception.

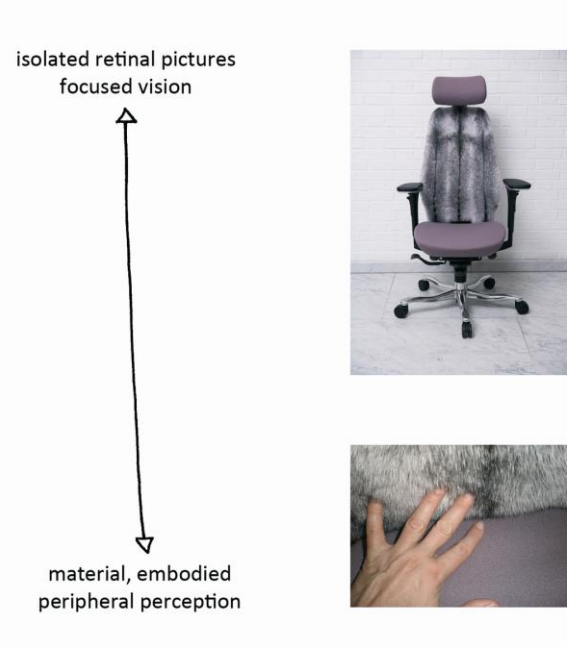


Figure 3. The figure shows Juhani Pallasmaa's concept of focused vision as opposed to peripheral perception, exemplified by different 'perceptions' of an office chair.

A HOLISTIC FRAMEWORK

Whereas Riegl's three views can be interpreted as the position of the observer in relation to textile design, Pallasmaa's concern about a multisensory and participant approach to the experience of architecture can be interpreted in relation to the character of the perception. That is, instead of defining the character of a view as either haptic or optic, the character of the perception can be that of an observer or that of a participant within each of the three perspectives.

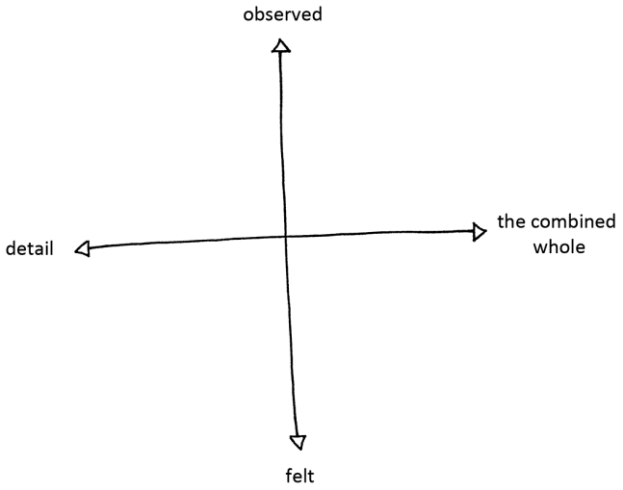


Figure 4. Alois Riegl's theory of perception is interpreted as a move from the detail to the combined whole. Juhani Pallasmaa's concept of focused vision and peripheral perception is interpreted as taking an observant or a participant perspective to design of applied textiles, i.e. from vision to multisensory perception.

The figure above suggests how to combine the position in relation to the object with the perception of the object. In textile design Riegl's three views can be interpreted as a move between perceiving the detail or the whole, that is, visual perception. Pallasmaa's focused vision and peripheral perception can be interpreted as a movement between being an observer or a participant, i.e. from vision to multisensory perception.

THE TRIPOD APPROACH

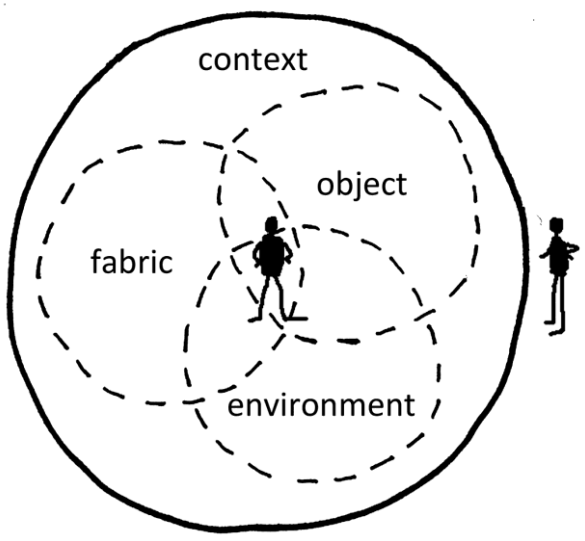


Figure 5. The figure shows a basic version of the Tripod Approach a structured and application-oriented framework for designing and exploring applied textiles.

The figure above captures three broad perspectives of textiles within a context. The Tripod Approach is based on three broad perspectives on textiles, i.e.

'textiles as materials', 'textiles as part of an object' and 'textiles as part of an environment'. The perspectives relate to 'an overall context'. The people in the figure indicate that the character of the perception can be based on either an observer's or a participant's perspective. The observer and the participant can represent the designer and the stakeholder, respectively, but it can also indicate that both designer and stakeholder can choose to perceive the textile from an observant as well as a participant perspective. Each view is encircled by a dotted line, which illustrates that it is flexible and can be defined broadly or narrowly depending on the actual context. The three perspectives are encircled by a solid line, which illustrates the importance of adapting the framework to a specific context.

The Tripod Approach can be used in two ways: 1) as a design tool for focusing on relevant and available means when designing future textiles and 2) as an analytical tool to articulate textile attributes. In other words, the framework can be used either bottom-up or top-down, i.e. as a tool for synthesis or for analysis. Another bottom-up and top-down perspective is that both synthesis and analysis can take their starting point in either the fabric appearance or the use context.

According to the Concise Oxford English Dictionary a 'tripod' is: "a three-legged stand for supporting a camera or other apparatus" or "a stool, table, or cauldron resting on three legs". In this context the word tripod is used figuratively in the sense that each leg symbolizes a perspective that represents the distance to the textile. The many possibilities for adjusting the tripod in order to obtain a stable support symbolize how the framework can be applied to a specific context. The operator's ability to control and apply the apparatus symbolizes that it is possible to approach the framework bottom-up or top-down as explained above. Finally the operator, whether he or she is controlling an apparatus or sitting on a stool, can symbolize the observer's versus the participant's perspective.

One intention is that the framework can capture applied textiles in a way that allows the designer as well as participating stakeholders to have 'a conversation with the material' (cp. Schön). The Tripod Approach embraces multiple entry levels in textile design. They are structured as three broad and flexible perspectives within an overall context. Thereby it is a systematic approach to articulating the multiple levels, which a textile designer has to consider when designing textiles. This systemized approach is assumed to be an advantage due to accessibility and operability since it encourages the textile designers to take advantage of the numerous possibilities they have to influence the textile design process, such as choice of yarn, technique, drape, functionality, purpose, style and so on. It is also seen as an advantage that the three perspectives remain flexible in order to ensure that the framework can be adapted to a specific context.

In this paper it is relevant to explore how the Tripod Approach contributes to designing for an emotional experience. Therefore the next section will give exemplify how the Tripod Approach can be adapted to the textile design process.

ADAPTING THE TRIPOD APPROACH TO THE TEXTILE DESIGN PROCESS

In this section an industrial textile design project exemplifies the use of the Tripod Approach. The project is conducted at a Danish textile manufacturer (for details see Bang, 2010). The 'Mesh' project represents a view of textile design as applied textiles designed for an object in an environment.

THE 'MESH' PROJECT

The 'Mesh' project was initiated when a customer expressed interest in a mesh solution for a backrest on an office chair. The idea was to develop a mesh backrest to an existing office chair, which originally was designed with an upholstered backrest. One customer requirement was to design a backrest that can be accepted on the contract market. Within textile design the contract market covers fabrics (and product solutions) for office furniture, theaters, public buildings etc. This means that the functional requirements such as abrasion resistance and color fastness to light are higher than for the home market.

At the Danish textile manufacturer an expressed customer interest leads to a market surveillance

exploring the potential for establishing a development project. Figure 6 visualizes how the Tripod Approach can support the identification of the overall context and three significant perspectives of the future product:

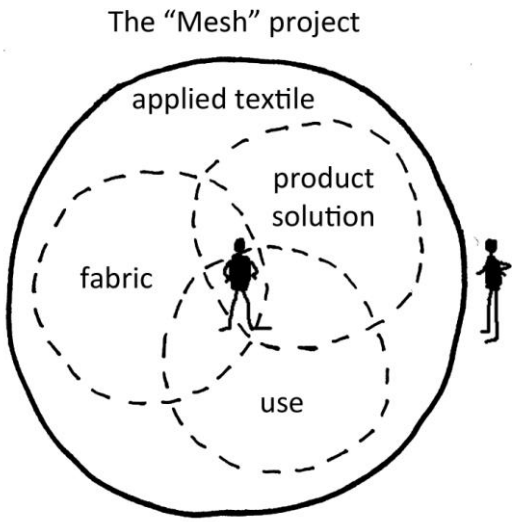


Figure 6. In the ‘Mesh’ project, where the overall context is ‘applied textiles’, the three perspectives are fabric, product solution and use.

First of all there are two clear entrance levels for the textile designer. One level is the fabric. Here the question is if it is necessary to develop a new mesh fabric or if an existing fabric can serve the purpose. The other clear entrance level is the backrest. This perspective is concerned with the product solution. The textile designer need to explore to what extent it is possible to influence for example the mounting of the mesh to the chair frame.

The third perspective in the Tripod Approach – the environment – is defined as ‘use’ in this case. The backrest is designed for an existing office chair, which means that ‘use’ is an inevitable parameter in the ‘Mesh’ project. Since it is a collaborative project between the customer and the textile manufacturer it is out of the hands of the textile designer to influence the use situation deeply. However knowledge about the use situation is valuable. Therefore the fact that the fabric or product solution is meant for the contract market is crucial information, which can be used to sharpen the design suggestions in the other entrance levels. Finally the three perspectives and entry levels in the design process – the fabric, the product solution and the use situation – is embedded in the overall context, which is expressed as a strong focus on the application of the textile design. This is seen as opposed to piece goods, where the fabric is produced for a market segment such as the contract or home market, but not necessarily for an intended use.

Finally there is the observant versus the participant perspective. In this case an observant perspective results in the market surveillance identifying the potential in a project like this. On the other hand there is also a very strong participant perspective in this project since it is conducted in collaboration with the customer who originally expressed the design interest.

Concerning using the Tripod Approach as a framework for designing for experience/emotional value I will return to one of the defined entrance levels – the fabric – with an example drawn from an interview with the textile designer. During the interview she explains how she decided to explore the customer’s design interest introducing mesh fabric as widely as possible.



Figure 7. The textile designer explores the customer’s design interest as widely as possible though textile samples.

After the introduction to the customer, which consisted of various fabric samples and combinations of fabrics three design stories were developed. The three design stories took different perspectives on mesh spanning from fabrics expressing the ‘idea of mesh’ to actual mesh fabrics.

The ‘Mesh’ project exemplify how flexibly the perspectives in the Tripod Approach can be interpreted according to the actual context, and still function as a structured and systematic approach to applied textiles securing a holistic view on the design process. Depending on the context the material

perspective can include levels from fiber to finished fabric. The object perspective can include levels from fabric to furniture component or another object. Finally the environment perspective can include levels concerned with intended use within a market segment or end-use.

SUMMING UP

This paper has introduced the Tripod Approach as a framework that can support designing the stimuli, which can lead to personal experiences of emotional value in relation to applied textiles. It was exemplified by the 'Mesh' project conducted at a Danish textile manufacturer. I used the examples from the Mesh project to show how the Tripod Approach can be used to access the relevant level of entry in the textile design process focusing on appropriate design parameters, and support designing the textile as part of an overall context.

FURTHER WORK

In future research it would be valuable to document design projects where the Tripod Approach is applied. It is relevant to ask if it is possible to apply the Tripod Approach to interdisciplinary design processes, or situations that involve stakeholders or end-users. Another question that might be worth exploring is if the Tripod Approach can be adapted to other design domains than textile design.

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